

Investigator Brochure: ArthroBalance-X

1. PHARMACOKINETICS & MECHANISM

ArthroBalance-X is a novel bispecific monoclonal antibody that provides dual immunomodulation for rheumatoid arthritis through simultaneous targeting of two key inflammatory pathways.

Mechanism 1: Selective inhibition of Tumor Necrosis Factor-alpha (TNF- α) receptor signaling with 8-fold greater affinity for membrane-bound TNF- α compared to soluble forms, preferentially affecting tissue-resident inflammation.

Mechanism 2: Simultaneous engagement of IL-6 co-receptor signaling on B cells and plasma cells, suppressing pro-inflammatory cytokine production while preserving regulatory T cell function.

2. DOSING & ADMINISTRATION

ArthroBalance-X is administered as a subcutaneous injection every 2 weeks at a fixed dose of 162mg. Dose escalation is not required. Self-injection training is provided to all patients. Peak serum concentration (C_{max}) is achieved in 3-4 days post-injection. No dose adjustment is required for mild hepatic impairment or mild renal impairment.

3. STORAGE & SHELF LIFE

ArthroBalance-X pre-filled syringes must be stored in a refrigerator at 2°C to 8°C in original cartons to protect from light. Temperature Warning: Storage at room temperature (20-25°C) for more than 8 hours or freezing will cause irreversible protein aggregation, rendering the product non-functional. Do not allow the product to freeze. Unopened syringes maintain stability for 24 months when stored at 2-8°C.